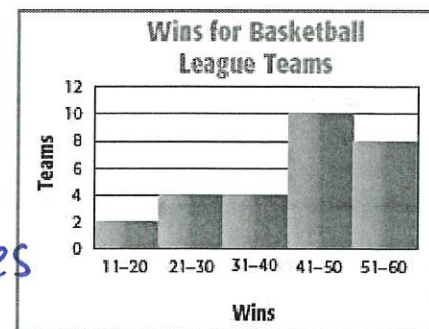
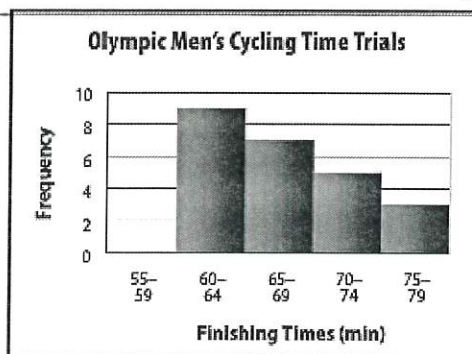


Histogram

A histogram is a type of bar graph used to display numerical data. The horizontal axis shows the range of the data separated into equal intervals with no gaps in between. The vertical axis shows the frequency of the data values. Each bar represents the # of data values within the specific range. How is a histogram different from a traditional bar graph? Displays numeric data



Example 1: Interpret data from a histogram



What is the range of data in each interval? 5

Which interval has 7 cyclists? 65-69

Which interval represents the greatest number of cyclists? 60-64

How many cyclists had a time less than 70 minutes? 16 = 9 + 7

How many total data values are there? 24

Example 2: Create a histogram for a given set of data.

Step 1: Create a frequency table to organize the data.

The intervals in the frequency table will be the intervals for the horizontal axis. 100-400

Daily Visitors to Selected State Parks				
108	209	171	152	236
165	244	263	212	161
327	185	192	226	137
193	235	207	382	241

Daily Visitors to Selected State Parks		
Visitors	Tally	Frequency
100-149	II	2
150-199	IIIT	7
200-249	IIIT	8
250-299	I	1
300-349	I	1
350-399	I	1

Step 2:

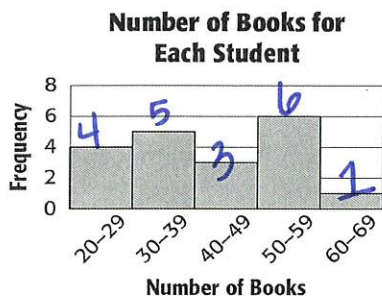
- Draw and label a horizontal and vertical axis.
- Write the title for the graph.
- Label intervals on horizontal axis.
- Label frequencies on vertical axis.

Step 3:

- Create a bar for each interval.



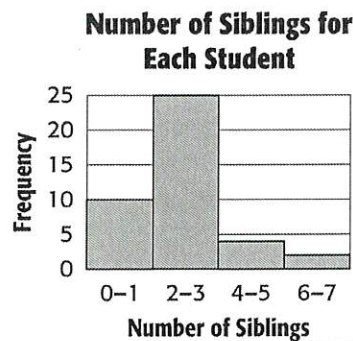
1. Give a set of data that could be represented by the histogram below.



Books	Frequency
20-29	4
30-39	5
40-49	3
50-59	6
60-69	1

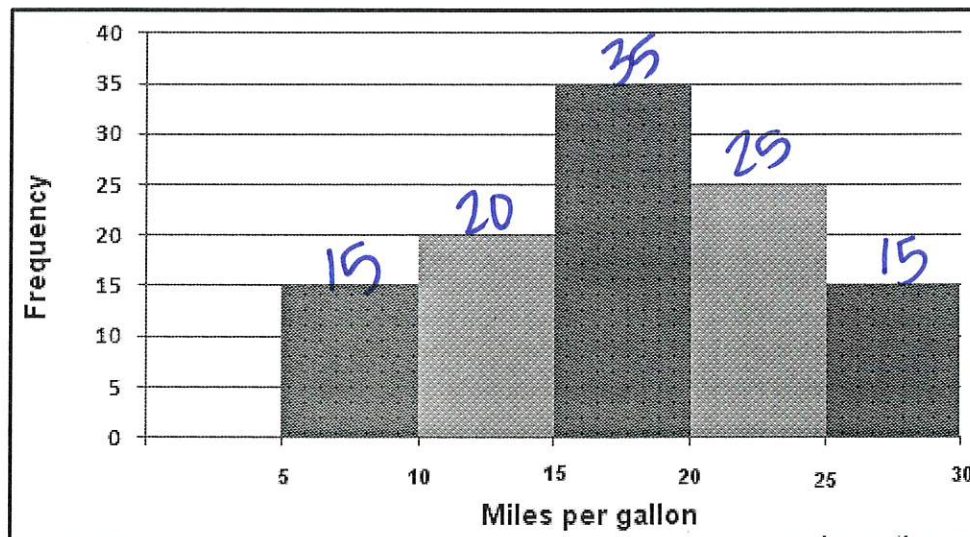
- i. How many students checked out 50 or more books? 7
- ii. How many total students are represented in the histogram? 19

2. While referring to the histogram below, decide if the statements below are true or false.



☐ T / ☐ F	Fewer than 5 students have 6 or more siblings.
☐ T / ☒ F	More students have 4 or more siblings than fewer than 4.
☐ T / ☐ F	Most students have 2 or 3 siblings.
☐ T / ☒ F	Every student has at least 1 sibling.

3. The histogram below shows the fuel consumption (miles per gallon) of a number of trucks.



- i. How many trucks had a fuel consumption less than 10mpg? 15 trucks
- ii. How many trucks had a fuel consumption between 10mpg and 20mpg? 55 trucks
- iii. Are there any gaps? Y/N Where are they? No
- iv. In what fuel consumption range are the most trucks? 15-20 mpg
- v. How many trucks had a fuel consumption at least 20 mpg? 40 trucks
≥ 20